

Session 4: Software Development Lifecycle (SDLC).

Task 1: The 6 Phases of SDLC (in Order)

1. **Requirements Gathering & Analysis:** Gathering detailed business needs and user requirements from stakeholders to plan what the software must achieve.
2. **Design:** Creating the technical blueprints, architecture diagrams, and user interface layouts based on the requirements.
3. **Development (Coding):** The phase where programmers write the actual source code to build the software based on the design specifications.
4. **Testing:** Evaluating the developed software to find bugs, fix errors, and ensure it meets all quality standards.
5. **Deployment:** Packaging and releasing the completed software to the production environment so real users can access it.
6. **Maintenance:** Monitoring the live software, fixing newly discovered bugs, and adding updates or improvements over time.

Task 2: Agile vs. Waterfall Model

Feature	Waterfall Model	Agile Model
Approach	Follows a strict, linear sequential path where one phase must finish before the next begins.	Follows an iterative and incremental path, developing software in small, continuous cycles.
Flexibility	Highly rigid; making changes after the project starts is difficult and expensive.	Highly flexible; adapts easily to changing requirements and user feedback at any stage.
Delivery	The complete software product is delivered only at the very end of the lifecycle.	Working software features are delivered frequently throughout the project lifecycle.

Task 3: The 3 Main Scrum Roles

- **Product Owner (PO):** Represents the business and customer needs, managing the

project requirements list (Product Backlog) to maximize product value.

- **Scrum Master:** Acts as a coach who facilitates team meetings, clears obstacles (blockers), and ensures the team follows Agile principles.
- **Development Team:** A self-organizing group of professionals (developers, designers, testers) who build and deliver the actual working software.

Task 4: Sprints in Scrum

What is a Sprint?

A Sprint is a fixed, time-boxed period during which a Scrum team works to complete a specific, committed set of tasks and features.

Typical Duration: A sprint typically lasts between **1 to 4 weeks**, with **2 weeks** being the most common length in the software industry.

Task 5: Jira Board Columns Explained

- **Backlog:** A holding area for all project tasks, feature ideas, and bugs that need to be done eventually, but are not yet scheduled for active work.
- **To Do:** Tasks that have been selected for the current cycle (Sprint) but have not yet been started by any team member.
- **In Progress:** Work items that a team member is actively working on, coding, or designing right now.
- **Done:** Completed tasks that have been successfully built, thoroughly tested, and meet the team's definition of final quality.